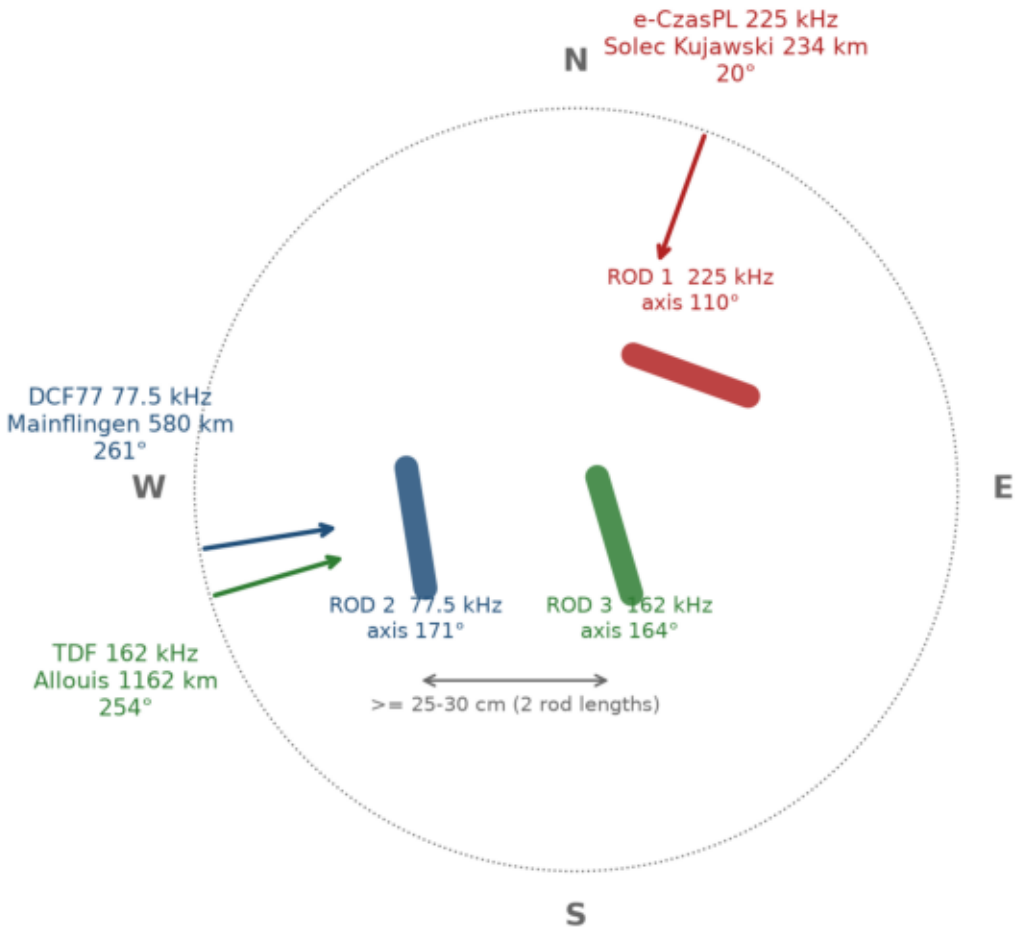
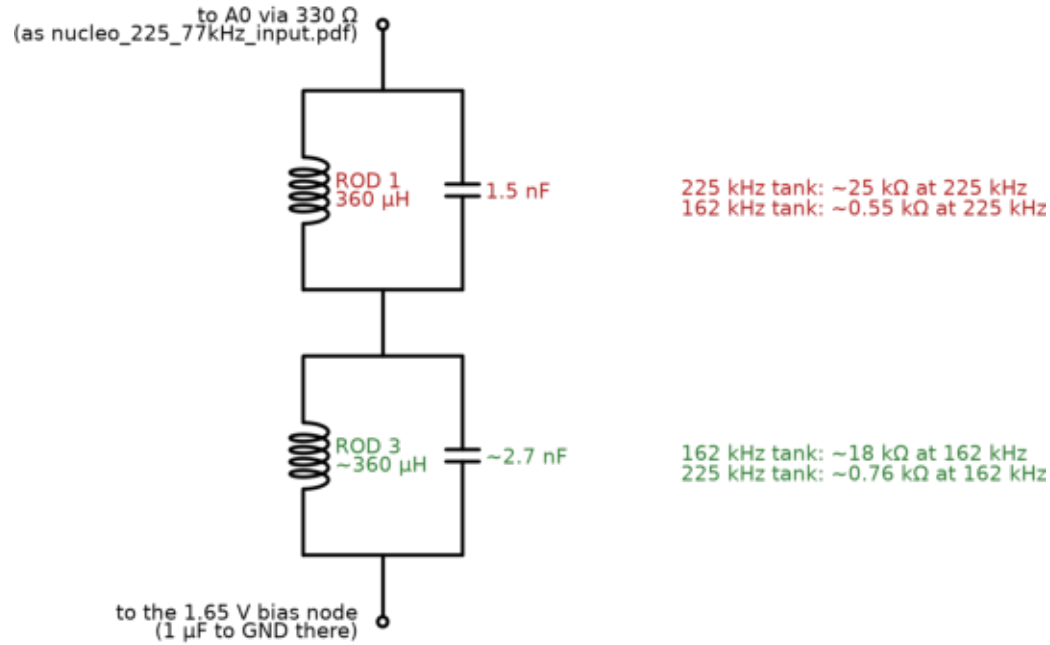
Three ferrite rods for e-CzasPL (225 kHz), TDF (162 kHz) and DCF77 (77.5 kHz)

Layout by station bearing from Wrocław (left) and the 225 + 162 kHz 'stacked tanks' on A0 (right); DCF77 stays on A1.



TOP VIEW, north up. A rod receives broadside (null along its axis): axis = bearing + 90°.
DCF77 and TDF are 7° apart -> their rods are almost parallel; e-CzasPL's is ~60° off them.



1. Not a triangle: an equilateral triangle would turn one of the two western rods 60° away from its station (-6 dB). Use the 'V': ROD 2 and ROD 3 ~N-S, ROD 1 ~ESE-WNW.
2. Parallel rods (ROD 2, ROD 3) couple most: >= 2 rod lengths apart side by side, or on one line end to end with a gap >= 1 rod length. ROD 1 at ~60°: 20-30 cm is enough.
3. Tuning capacitors near the board (as before). Retune each rod with the others in place - the TDF probe build (-DF_DCF=162000) shows C/N0 within minutes.
4. ROD 3: measure its inductance first; C = 1 / (4 pi² f² L): 360 µH -> 2.68 nF at 162 kHz (e.g. 2.2 nF + 470 pF, then trim). Values with ~ are estimates.
5. Stacked tanks: each station's voltage reaches A0 with ~3-4 % loss; the ADC samples 0-500 kHz, the firmware separates the stations (mixers).